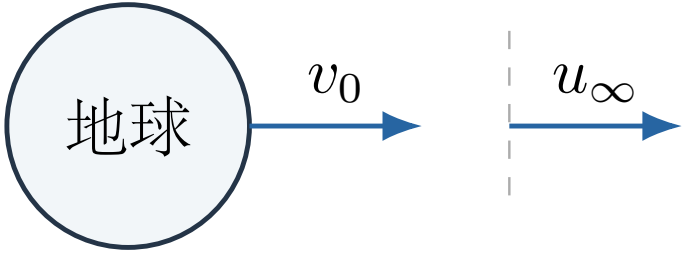
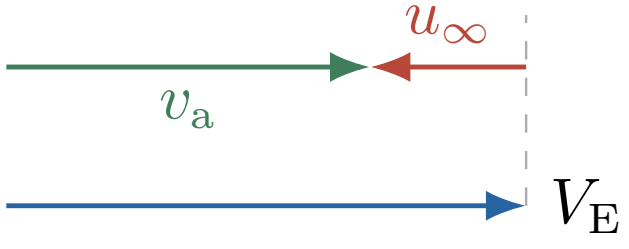


$$\frac{v_0^2}{2} - \frac{GM_\mathrm{E}}{r_\mathrm{E}} = \frac{u_\infty^2}{2}$$



$$v_\mathrm{a} = V_\mathrm{E} - u_\infty$$